

Lesson 2.1: Electronic Components and Circuit Design — Part 1

Electronics & Physical Computing

Passive Components

What are Passive Electronic Components?

Passive electronic components are components that do not require an external power source to operate and **cannot amplify signals**. They **consume or store energy**, rather than generate it. These components are essential for controlling current, filtering signals, and storing energy in circuits.

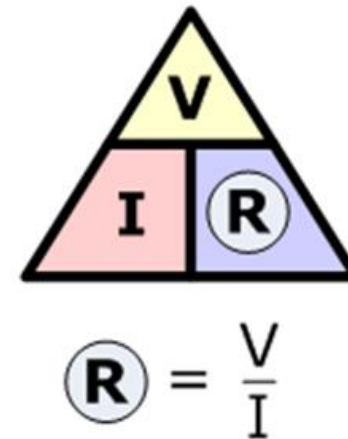
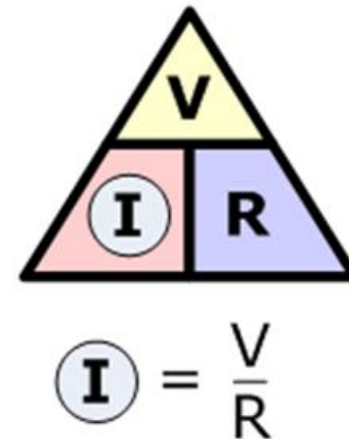
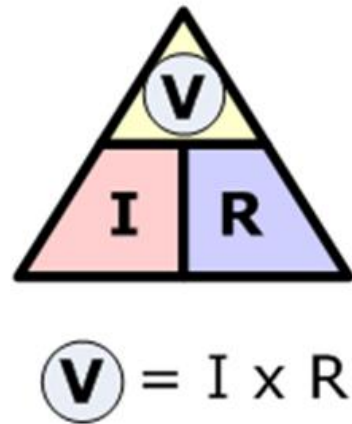
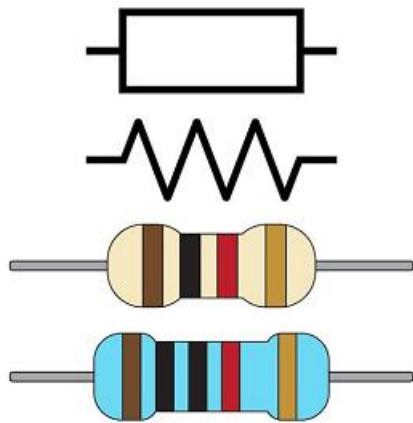


Resistor (R)

What is a Resistor?

A **resistor** is a passive component that **opposes or limits the flow of electric current** in a circuit. It does this by converting electrical energy into heat. Resistors are essential for controlling voltage and current in electronic devices.

The **unit** of resistance is the **ohm (Ω)** - Named after **Georg Simon Ohm**, who formulated **Ohm's Law**.



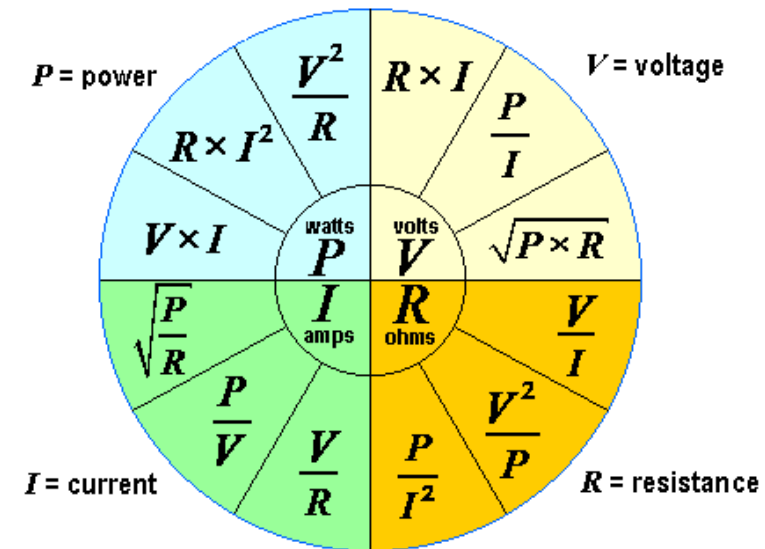
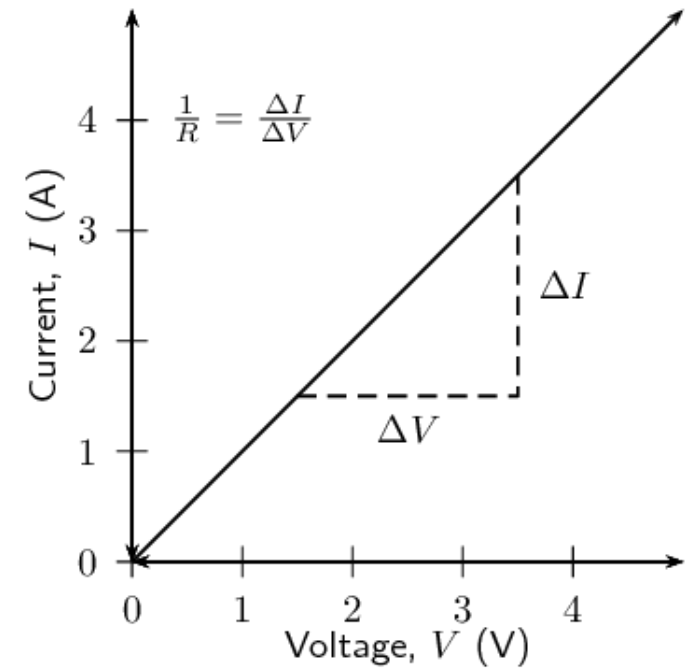
Ohm's Law

The relationship between **voltage (V)**, **current (I)**, and **resistance (R)** is given by:

$$V = I \times R$$

This means:

- If you increase resistance while keeping voltage constant, current decreases.
- If resistance is zero, current flows freely (short circuit).
- If resistance is infinite, no current flows (open circuit).

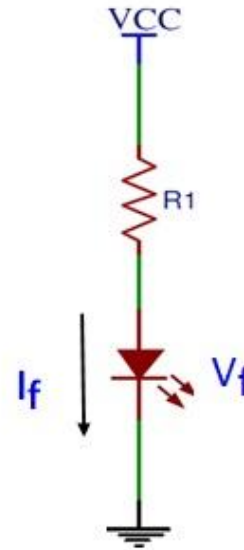
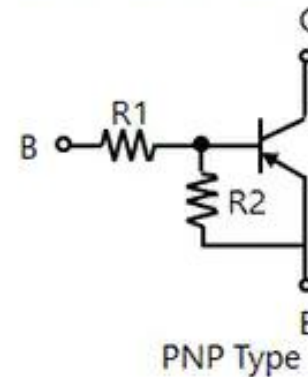
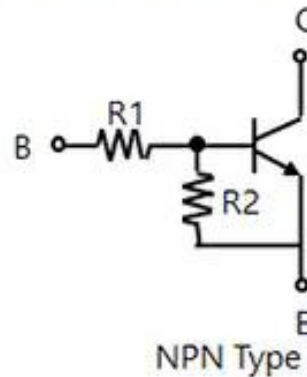
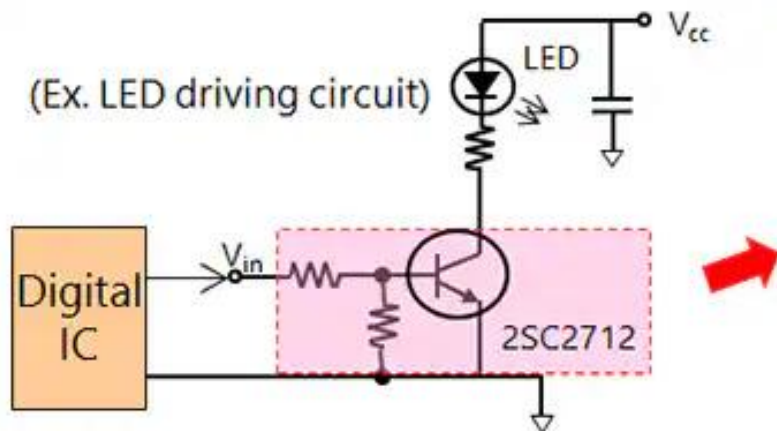




<https://www.youtube.com/watch?v=DYcLFHgVCn0>

Purpose of Resistors

- **Limit current** to prevent damage to components.
- **Divide voltage** in voltage divider circuits.
- **Bias active components** like transistors.
- **Pull-up or pull-down** logic pins in digital circuits.
- **Control timing** when used with capacitors (e.g., RC circuits).



By Ohm's Law

$$R = \frac{V_{cc} - V_f}{I_f}$$

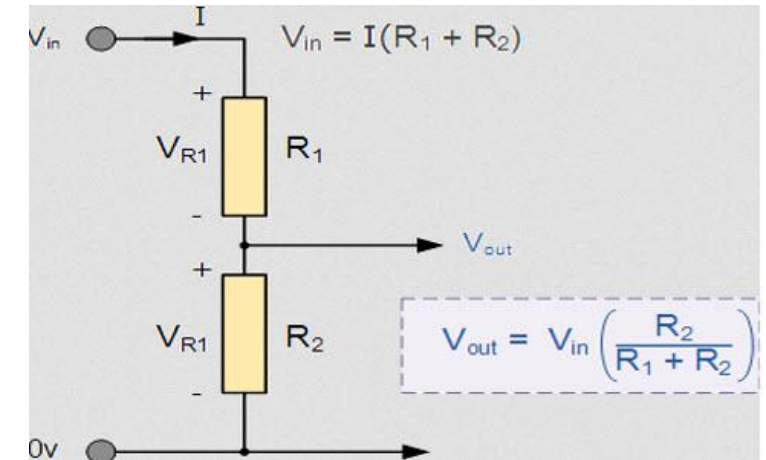
V_{cc} = Input Voltage

V_f = Forward Voltage of LED

I_f = Forward Current of LED

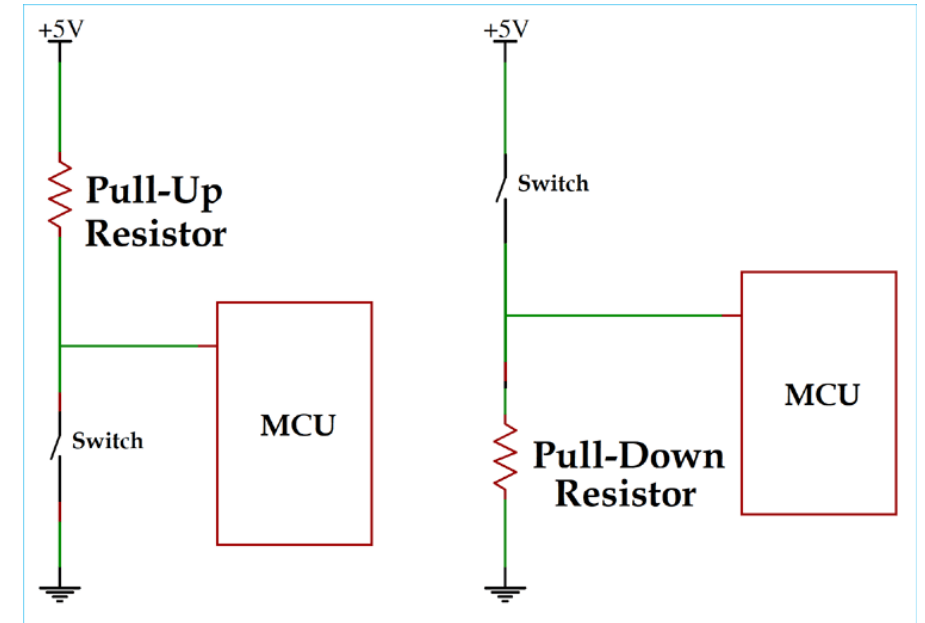
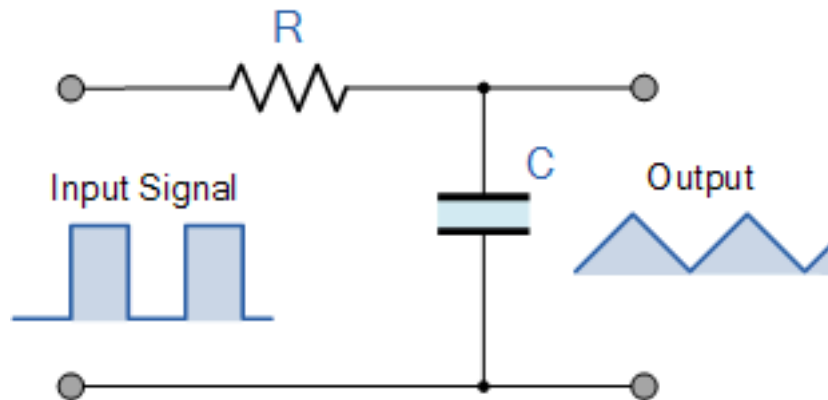
R = Resistance

Note: V_f = Minimum voltage required to light up an LED. It depends on color of an LED



Purpose of Resistors Cont.

- **Limit current** to prevent damage to components.
- **Divide voltage** in voltage divider circuits.
- **Bias active components** like transistors.
- **Pull-up or pull-down** logic pins in digital circuits.
- **Control timing** when used with capacitors (e.g., RC circuits).



Reading Resistor Values

Color Code System (4-band example):

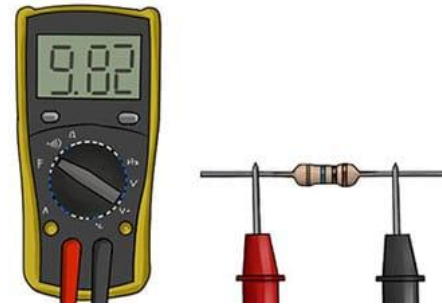
Each resistor has colored bands indicating its resistance value. For 4-band resistors:

- Band 1 & 2: Significant digits
- Band 3: Multiplier
- Band 4: Tolerance (accuracy)

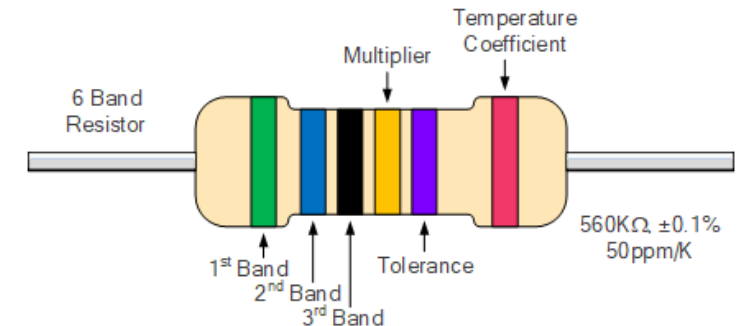
Example:

- Brown (1), Black (0), Red ($\times 100$), Gold ($\pm 5\%$) $\rightarrow 1,000 \Omega$ or $1 \text{ k}\Omega \pm 5\%$

Or you can use a **multimeter** to measure it directly.



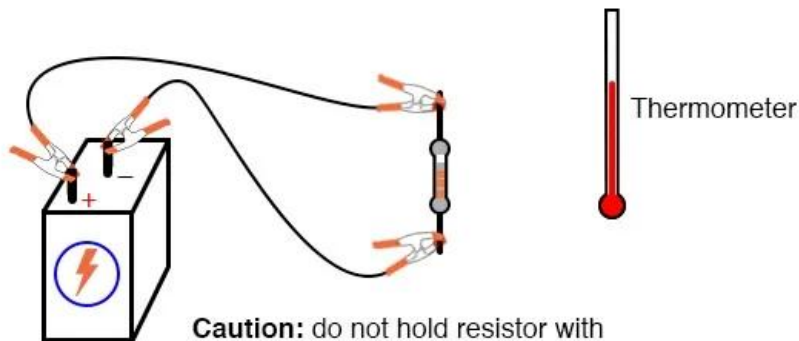
	1 st Digit	2 nd Digit	Multiplier	Tolerance
Black	0	0	$\times 1$	
Brown	1	1	$\times 10$	$\pm 1\%$
Red	2	2	$\times 10^2$	$\pm 2\%$
Orange	3	3	$\times 10^3$	$\pm 3\%$
Yellow	4	4	$\times 10^4$	$\pm 4\%$
Green	5	5	$\times 10^5$	$\pm 0.5\%$
Blue	6	6	$\times 10^6$	$\pm 0.25\%$
Violet	7	7	$\times 10^7$	$\pm 0.1\%$
Grey	8	8	$\times 10^8$	$\pm 0.05\%$
White	9	9	$\times 10^9$	
Gold			$\times 10^{-1}$	$\pm 5\%$
Silver			$\times 10^{-2}$	$\pm 10\%$



Power Rating

- Measured in **watts (W)**.
- Indicates how much power the resistor can safely dissipate as heat.
- Common ratings: 1/8 W, 1/4 W, 1/2 W, 1 W, 2 W
- Exceeding this limit can **burn** the resistor.

$$\text{Power (P)} = V \times I = I^2 \times R = \frac{V^2}{R}$$



Caution: do not hold resistor with your fingers while powered!

1/4 Watt



1/2 Watt



1 Watt



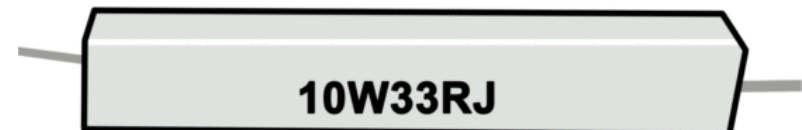
2 Watt



5 Watt

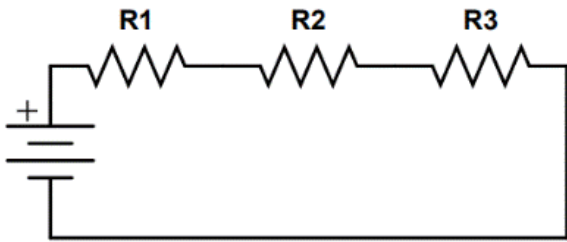


10 Watt

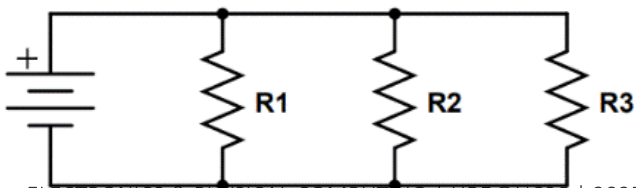


Ways to Connect Resistors: Series Connection

Resistors in Series



Resistors in Parallel



- **Connection:** End-to-end, one after another.
- **Total Resistance (R_t):**

$$R_t = R_1 + R_2 + R_3 + \dots$$

- **Current:** Same through all resistors.
- **Voltage:** Divided among resistors.

✓ Applications:

- **Voltage dividers:** Used to get a specific voltage from a higher source.
- **LED protection:** Sometimes used to spread heat dissipation.

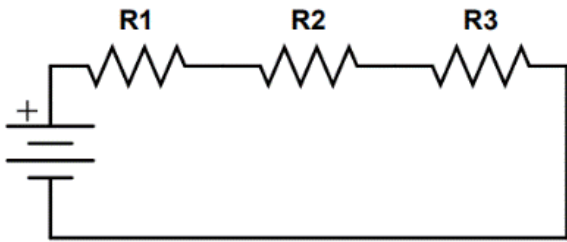
🔧 Example:

If you have 3 resistors: 1 k Ω , 2 k Ω , and 3 k Ω in series:

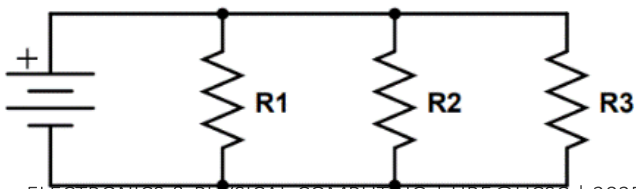
$$R_t = 1k + 2k + 3k = 6k\Omega$$

Ways to Connect Resistors: Parallel Connection

Resistors in Series



Resistors in Parallel



ELECTRONICS & PHYSICAL COMPUTING | HBE@UCSC | 2025

- **Connection:** All resistor ends are connected to the same two points.
- **Total Resistance (R_t):**

$$\frac{1}{R_t} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots$$

- **Voltage:** Same across all resistors.
- **Current:** Divides between resistors.

✓ Applications:

- Reducing overall resistance
- Sharing current load
- Creating fault-tolerant paths (if one fails, others still conduct)

🔧 Example:

If you have two 1 k Ω resistors in parallel:

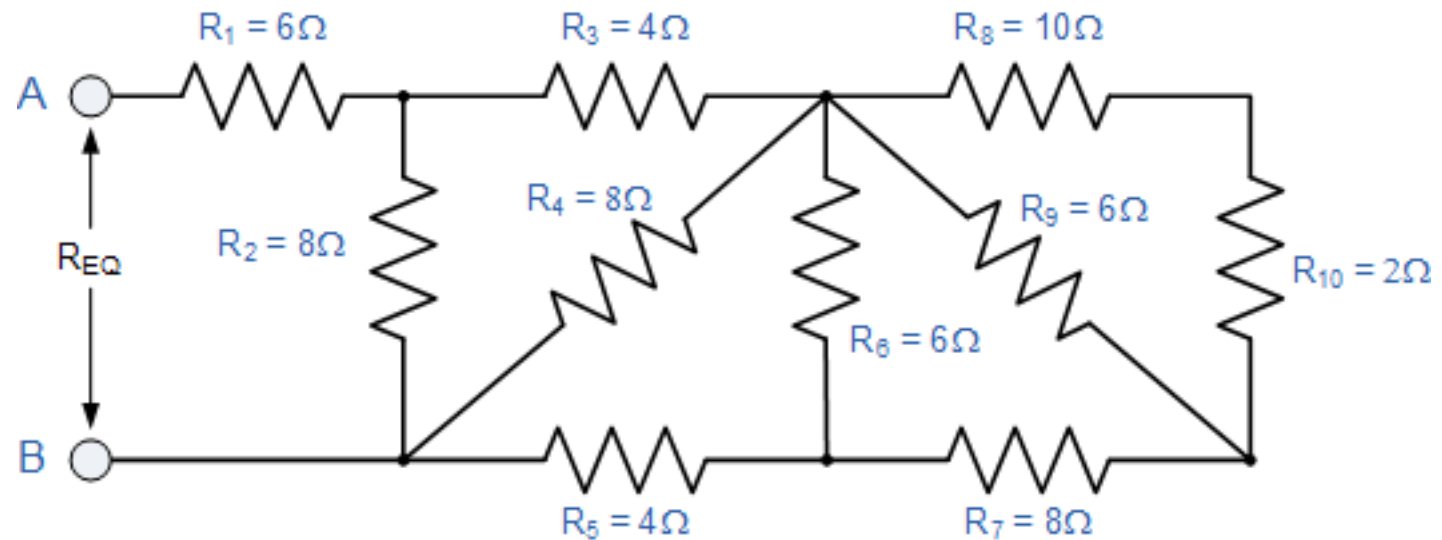
$$\frac{1}{R_t} = \frac{1}{1000} + \frac{1}{1000} = \frac{2}{1000} \Rightarrow R_t = 500\Omega$$

Ways to Connect Resistors: Series-Parallel Combination

- Mix of series and parallel resistors.
- Useful for **design flexibility** and achieving specific resistance values.

✓ Applications:

- Power distribution networks
- Custom voltage and current control circuits



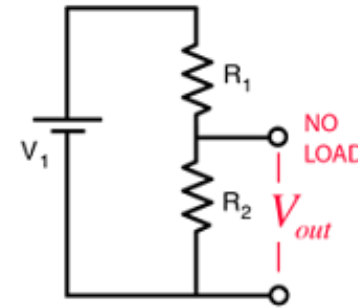
Real-Life Applications of Resistor Connections

Voltage Divider (Series)

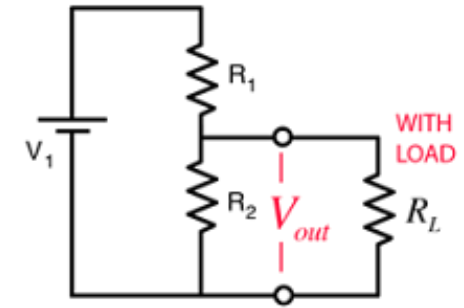
- Two resistors in series across a voltage source
- Output voltage taken between them

$$V_{out} = V_{in} \times \frac{R_2}{R_1 + R_2}$$

OPEN CIRCUIT BEHAVIOR



BEHAVIOR UNDER LOAD



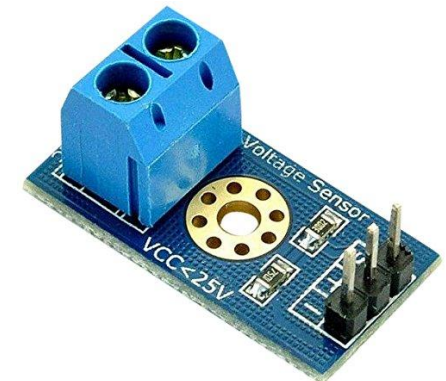
$$V_{out} = V_1 \frac{IR_2}{I(R_1 + R_2)} = \frac{V_1 R_2}{(R_1 + R_2)}$$

OUTPUT VOLTAGE UNDER
"NO LOAD" CONDITION
(open circuit)

OUTPUT VOLTAGE
UNDER LOAD

$$V_{out} = V_1 \frac{IR_2}{I(R_1 + R_2)} = \frac{V_1 (R_2 \parallel R_L)}{(R_1 + R_2 \parallel R_L)}$$

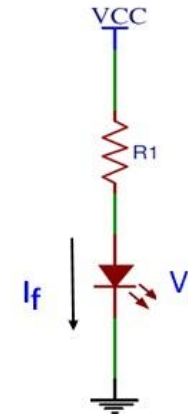
 **Application:** Reading analog voltage on Arduino (e.g., battery monitoring)



Real-Life Applications of Resistor Connections

🔧 Current Limiting (Series)

- A resistor is placed in series with a component like an **LED** to limit current.



By Ohm's Law

$$R = \frac{V_{cc} - V_f}{I_f}$$

V_{cc} = Input Voltage

V_f = Forward Voltage of LED

I_f = Forward Current of LED

R = Resistance

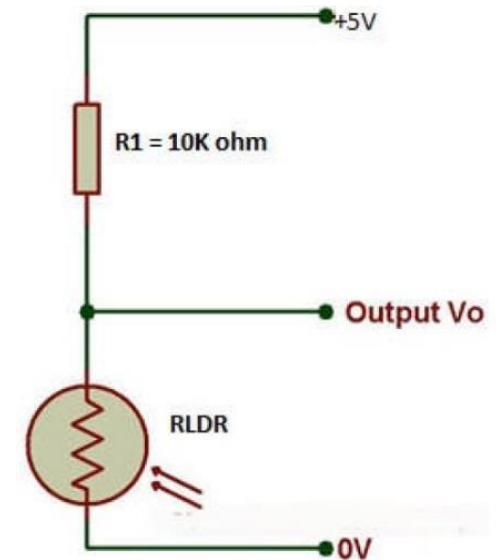
Note: V_f = Minimum voltage required to light up an LED. It depends on color of an LED

✅ **Application:** LED or buzzer protection

🌡️ Sensors with Resistors (Voltage Divider + Analog Read)

- Thermistors or LDRs are used in a **voltage divider** to convert variable resistance into voltage.

✅ **Application:** Light sensors, temperature monitoring, etc.



Real-Life Applications of Resistor Connections

Load Sharing (Parallel)

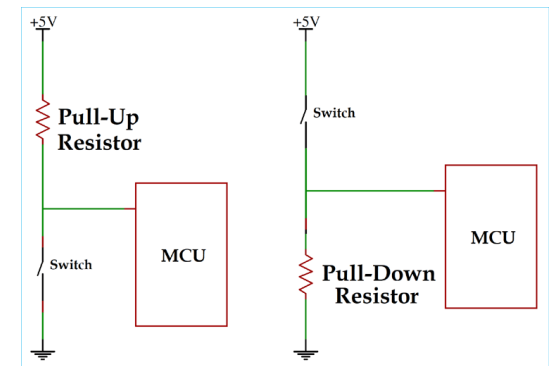
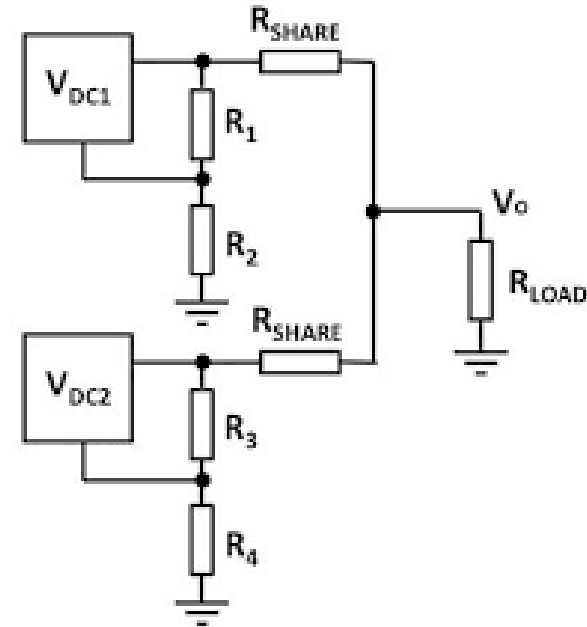
- Multiple resistors in parallel to **share power dissipation** or adjust resistance.

✓ **Application:** High-power circuits, DIY power resistors

Pull-up/Pull-down Resistors

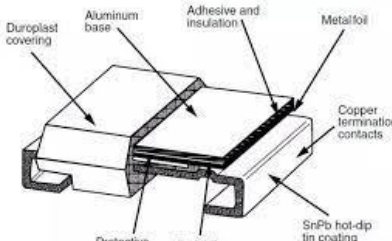
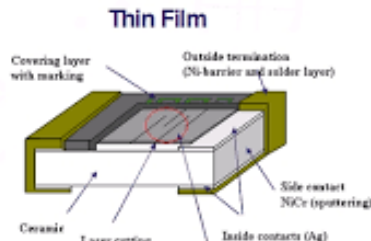
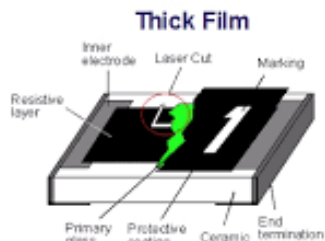
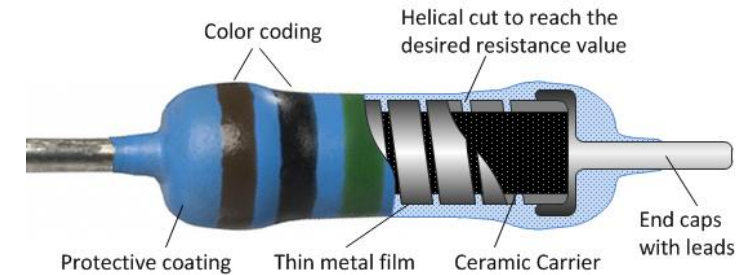
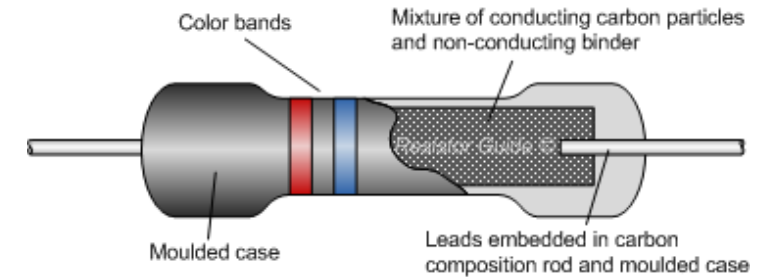
- Connected to **digital input pins** to set a known logic level when no input is present.

✓ **Application:** Microcontrollers like Arduino or Raspberry Pi

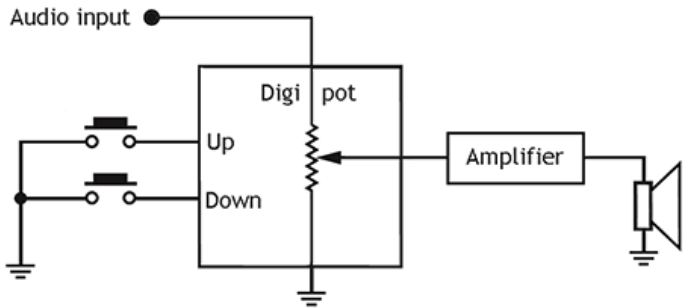


Types of Resistors: By Construction / Material

Type	Description	Use Case
Carbon Film	Carbon deposited on ceramic rod	General-purpose, cheap
Metal Film	Metal layer on ceramic rod	Precise, low noise
Wire-Wound	Wire wound around a core	High power, inductive
Thick/Thin Film	Printed on a substrate	Surface-mount devices (SMD)
Foil Resistors	Ultra-precise, low temperature coefficient	High-precision instruments

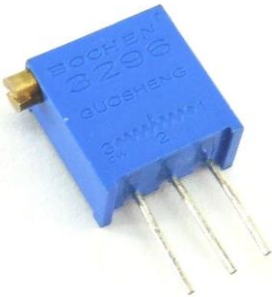
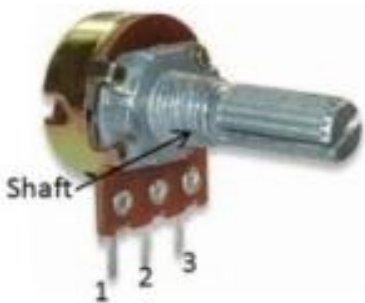
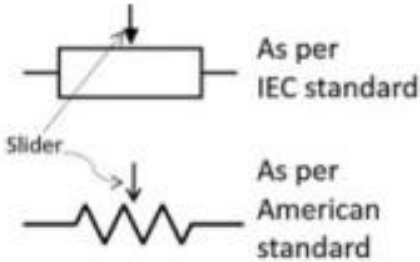
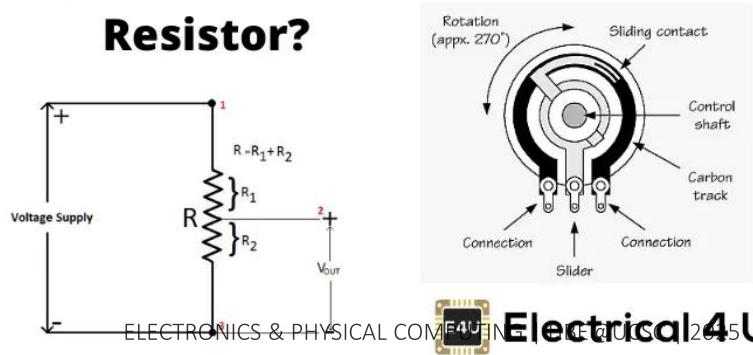


Types of Resistors: By Function

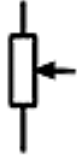
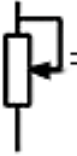
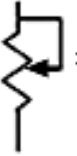


Type	Description	Use Case
Fixed Resistors	Constant resistance value	Most circuits
Variable Resistors (Potentiometers)	Adjustable manually or electronically	Volume knobs, tuning
Preset/Trimmer	Small adjustable resistors, set during calibration	Fine-tuning circuits
Digital Potentiometers	Electronically controlled resistance	MCU-controlled variable resistance

What is a Variable Resistor?



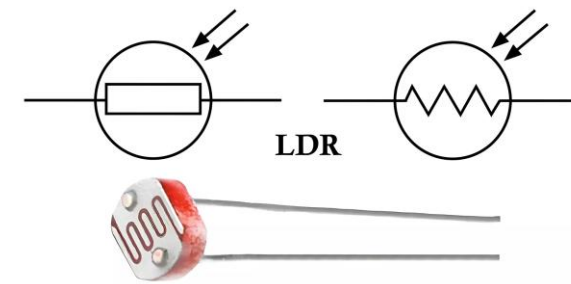
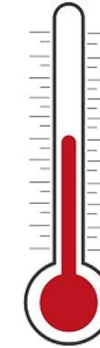
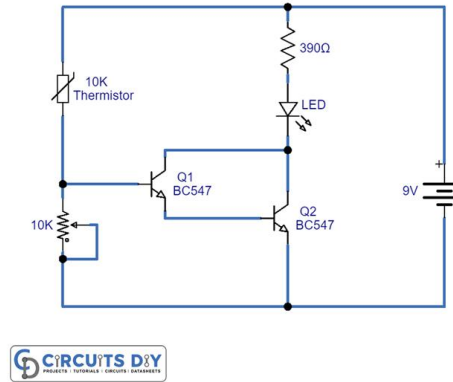
Variable Resistor, Potentiometer, and Preset

	European Symbols		US Symbols	
Potentiometers				
Variable Resistors				
	User Controls	Pre-set Controls	User Controls	Pre-set Controls

Term	Terminals	Function	Adjustment Style	Typical Use
Potentiometer	3	Variable voltage (voltage divider)	Knob or slider	User-adjustable controls (e.g., volume, brightness)
Variable Resistor	2 (wiper + 1 end)	Variable resistance (current control)	Knob, slider, or screw	Limiting current, tuning components
Preset (Trimmer)	2 or 3	Calibration or tuning	Screwdriver (infrequent)	Factory or service calibration

Types of Resistors: Special Behaviour Types

Temperature Sensor Circuit



Type	Description	Use Case
Thermistors	Resistance varies with temperature	Temperature sensors
- NTC	Resistance ↓ with temp ↑	Thermometers, inrush current limiters
- PTC	Resistance ↑ with temp ↑	Overcurrent protection
LDR (Photoresistor)	Resistance ↓ with light ↑	Light sensors, auto-brightness
Varistor (VDR)	Resistance ↓ sharply after a voltage threshold	Surge protection

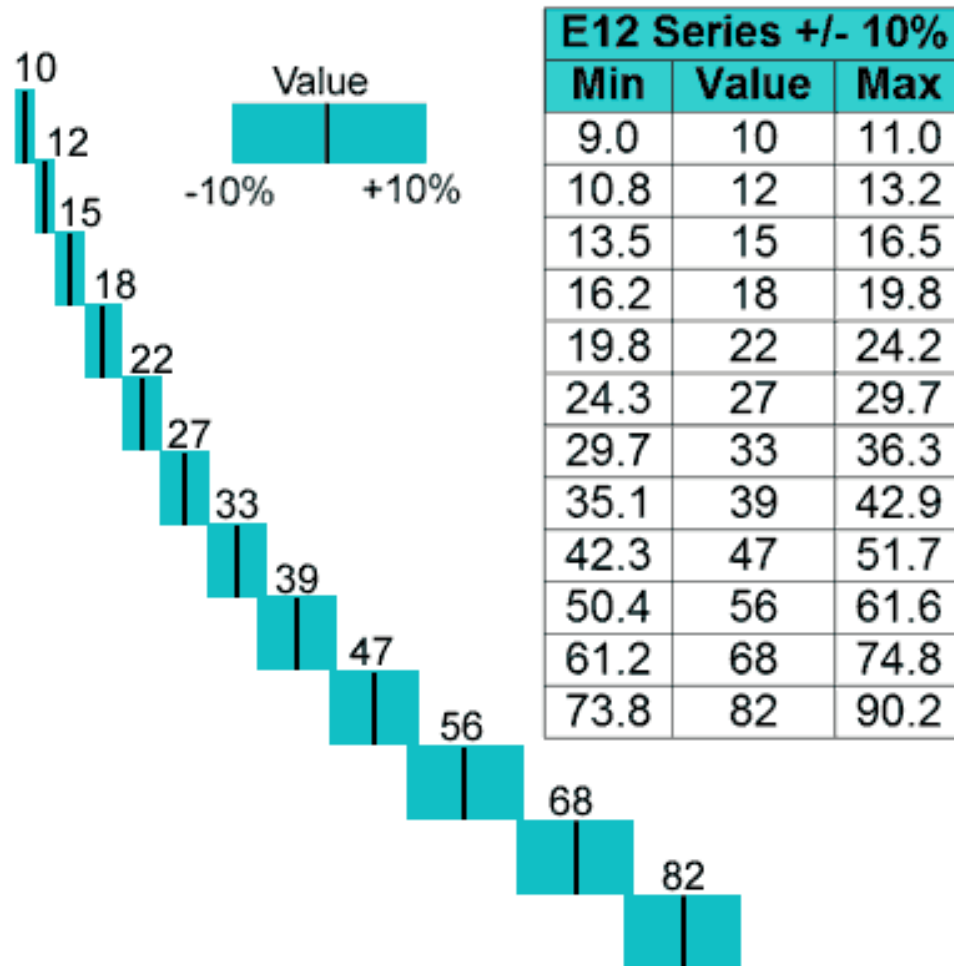
Resistor Standards

To ensure **interchangeability**, **manufacturing consistency**, and **predictable performance**, resistors are made following international standards.

Resistors come in standard values defined by **E-series** (E6, E12, E24, E48, E96, E192), which divide each decade (like 10–100, 100–1000 Ω) into specific steps.

Series	Tolerance	Common for
E6	$\pm 20\%$	General low-accuracy use
E12	$\pm 10\%$	Consumer electronics
E24	$\pm 5\%$	Standard fixed resistors
E96	$\pm 1\%$	Precision applications
E192	$\pm 0.5\%$, 0.1%	High-precision instruments

E.g. E12 Series



Example: E12 series values in a decade (10–100)

10, 12, 15, 18, 22, 27, 33, 39, 47, 56, 68, 82

Multiply by powers of 10 to get full range:

E.g., 10 Ω , 100 Ω , 1k Ω , 10k Ω , etc.

Other Standards

2. Tolerance

- Indicates how much the actual resistance may differ from the labeled value.
- Given in %:
e.g., $1\text{ k}\Omega \pm 5\%$ → Value can range from $950\ \Omega$ to $1050\ \Omega$.

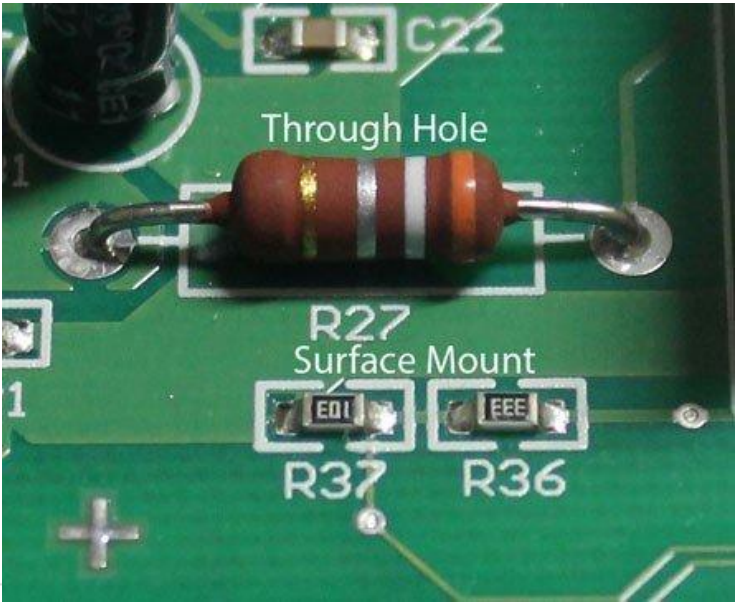
3. Power Rating

- Standard resistor power ratings: $1/8\text{W}$, $1/4\text{W}$, $1/2\text{W}$, 1W , 2W , 5W , etc.
- Choose a resistor with a power rating **at least 2x** the expected power dissipation.

Other Standards

4. Package Types

Type	Example	Description
Through-hole	1/4W axial	For breadboards and hobbyist circuits
SMD (Surface Mount)	0805, 0603	For compact, PCB-based designs



223

223
= 22×10^3
= 22,000 Ohm
= 22K Ohm

Three-Digit Resistor

4R7

4R7
= 4.7 Ohm

Resistor With Radix Point

0

0
= 0 Ohm

8202

8202
= 820×10^2 Ohm
= 82,000 Ohm
= 82 KOhm

Four-Digit Resistor

0R22

0R22
= 0.22 Ohm

Resistor With Radix Point

000

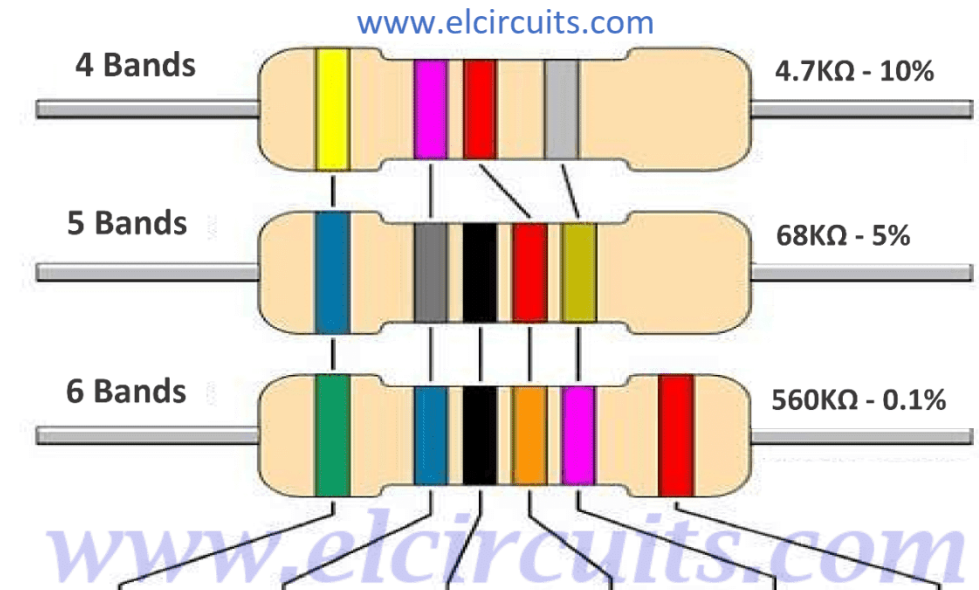
000
= 0 Ohm

Resistor Color Code - 4 Band, 5 Band and 6 Band

Exercise #1

Q1) A 4-band resistor has the color bands: **Yellow – Violet – Red – Gold**. Determine its resistance value. Determine the minimum and maximum possible resistance due to tolerance.

Q2) A 5-band precision resistor has the colors: **Brown – Green – Black – Red – Brown**. Calculate the exact resistance value and state its tolerance.



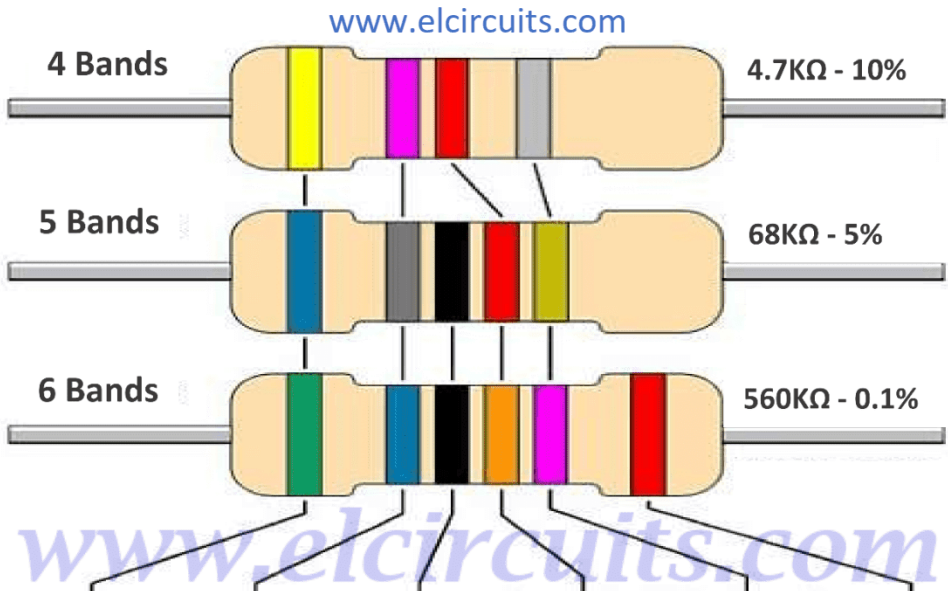
Color	1st Band	2nd Band	3rd Band	Multiplier	Tolerance	Temp. Coefficient
Black	0	0	0	X 1Ω		
Brown	1	1	1	X 10Ω	$\pm 1\%$	100ppm
Red	2	2	2	X 100Ω	$\pm 2\%$	50ppm
Orange	3	3	3	X $1\text{K}\Omega$		15ppm
Yellow	4	4	4	X $10\text{K}\Omega$		25ppm
Green	5	5	5	X $100\text{K}\Omega$	$\pm 0.5\%$	
Blue	6	6	6	X $1\text{M}\Omega$	$\pm 0.25\%$	
Violet	7	7	7	X $10\text{M}\Omega$	$\pm 0.1\%$	
Grey	8	8	8		$\pm 0.05\%$	
White	9	9	9			
Gold				X 0.1Ω	$\pm 5\%$	
Silver				X 0.01Ω	$\pm 10\%$	

Exercise #2

Q3) You have a resistor labeled $2.7\text{ k}\Omega \pm 5\%$. What 4-band color code corresponds to this resistor?

Q4) A resistor with the colors **Orange – Orange – Black – Brown** is installed in a circuit. What is the nominal resistance? What is the range of resistance values allowed by the tolerance?

Resistor Color Code - 4 Band, 5 Band and 6 Band



Color	1st Band	2nd Band	3rd Band	Multiplier	Tolerance	Temp. Coefficient
Black	0	0	0	X 1 Ω		
Brown	1	1	1	X 10 Ω	$\pm 1\%$	100ppm
Red	2	2	2	X 100 Ω	$\pm 2\%$	50ppm
Orange	3	3	3	X 1K Ω		15ppm
Yellow	4	4	4	X 10K Ω		25ppm
Green	5	5	5	X 100K Ω	$\pm 0.5\%$	
Blue	6	6	6	X 1M Ω	$\pm 0.25\%$	
Violet	7	7	7	X 10M Ω	$\pm 0.1\%$	
Grey	8	8	8		$\pm 0.05\%$	
White	9	9	9			
Gold				X 0.1 Ω	$\pm 5\%$	
Silver				X 0.01 Ω	$\pm 10\%$	

Answers

Q1) **4.7 k Ω $\pm$ 5%** (4,465 Ω to 4,935 Ω).

Q2) **15 k Ω $\pm$ 1%** (14,850 Ω to 15,150 Ω).

Q3) **Red – Violet – Red – Gold** (i.e., $2 - 7 - \times 10^2 - \pm 5\%$).

Q4) **33 Ω $\pm$ 1%** (32.67 Ω to 33.33 Ω).